Active query synthesis for preference learning

Georgia Institute of Technology

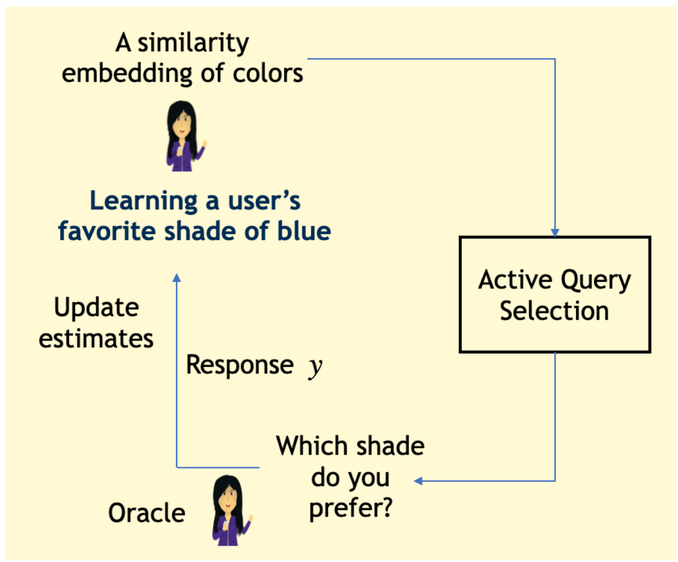
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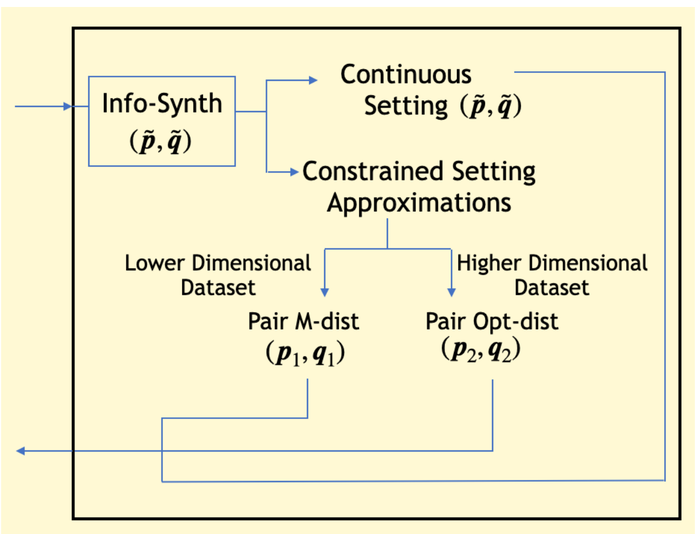
SCAN ME

Active learning

- Involves **adaptively** querying labels for the most informative data points to improve **label efficiency**
- But standard pool-based active learning methods are computationally **expensive**
- We introduce **Info-Synth**, an active query synthesis framework for continuous settings and two approximation methods - **Pair M-dist** and **Pair Opt-dist** - that extend query synthesis to fixed datasets



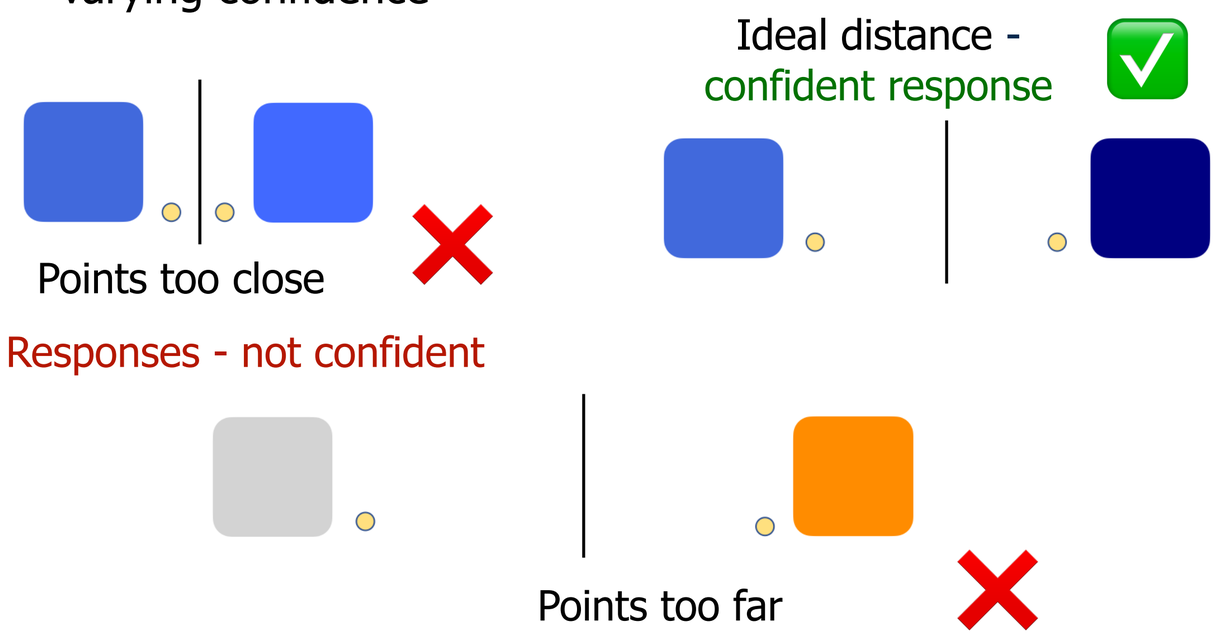
Active Learning framework



Active Query Selection

Learning human preferences

- Estimate a user preference vector $w \in \mathbb{R}^d$ based on binary preference responses y over pairs of items (p, q)
- Responses have different levels of associated confidence
- Existing response models generally fail to capture this varying confidence



Contributions

Confidence-aware response model

$$P(p \prec q) = \Phi\left(\frac{\|w-q\|^2 - \|w-p\|^2}{\sigma_0\sqrt{\|w-q\|^4 + \|w-p\|^4}}\right)$$

σ_0 represents the noise parameter
Φ represents the link function

Identifies queries with ideal distances

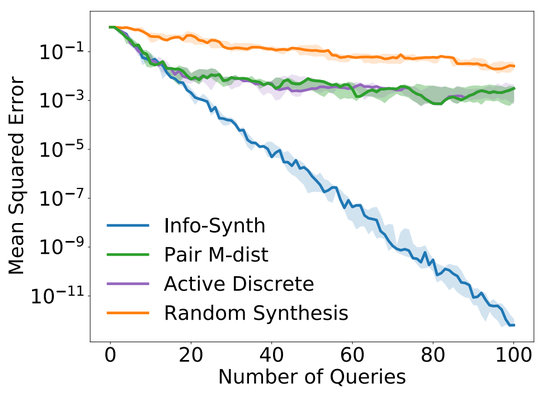
Info-Synth method

Maximization of mutual information

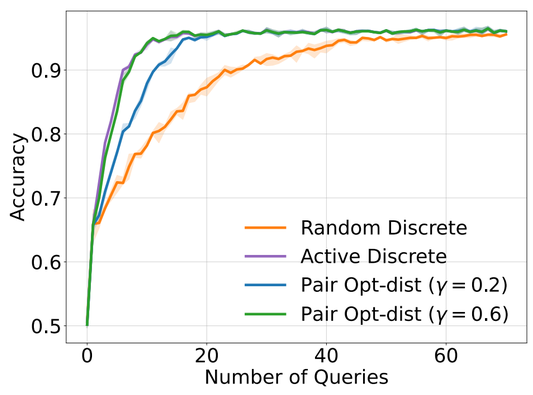
$$I\left(Y; W \mid (p,q)\right) = H\left(Y \mid (p,q)\right) - \mathop{\mathbb{E}}_{W}\left[H(Y \mid W, (p,q))\right]$$

$$(\tilde{p}, \tilde{q}) = \arg\max_{(p,q)} I\left(Y; W \mid (p,q)\right)$$

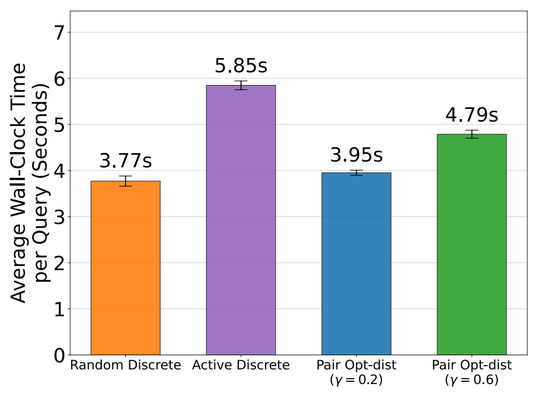
Experiment results

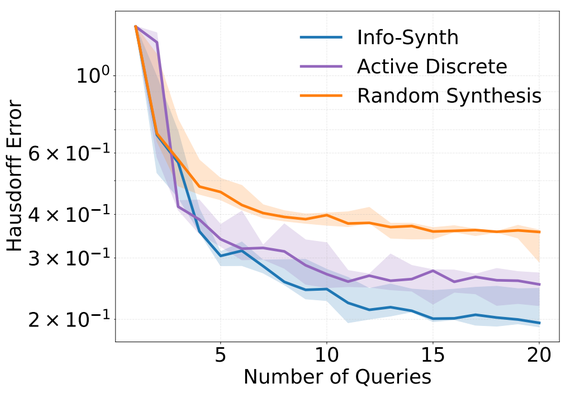


Synthetic Data



Reddit Summary Dataset





Robot Gain Tuning

Approximation methods

- Filter and refine strategies
  - filter the candidate queries based on minimizing approximate criteria
  - refine using pool-based MI evaluations
- **Pair M-dist**: filtering done using Mahalanobis distances for lower dimensional datasets
- **Pair Opt-dist**: filtering done using an optimality heuristic for higher dimensional datasets

Results

- **Synthetic Preference Learning:** *Info-Synth* outperforms baseline methods in continuous spaces, while the *Pair M-dist* strategy matches discrete (pool-based) method accuracy with higher computational efficiency
- **Reddit Summary Dataset:** *Pair Opt-dist* matches discrete (pool-based) baseline performance significantly faster, offering an adjustable parameter (γ) to balance computation speed and sample efficiency
- **Robot Gain Tuning:** *Info-Synth* successfully optimizes continuous controller parameters, achieving the lowest trajectory tracking error with minimal human feedback